

Chapter 1

INTRODUCTION

The healthcare industry faces significant challenges in managing the vast and ever-increasing amount of patient data. One critical area where this data can be particularly impactful is in drug recommendation, where healthcare specialists must analyze patient feedback and clinical data to suggest appropriate medications. The traditional approach to drug recommendation is often time-consuming and relies heavily on the expertise and availability of specialists, leading to potential delays in patient care and increased workload for healthcare professionals.

In recent years, advancements in machine learning (ML) and natural language processing (NLP) have opened new avenues for automating and enhancing drug recommendation processes. These technologies offer the potential to analyze large datasets efficiently, extracting meaningful insights from patient reviews and clinical reports. By leveraging these insights, we can develop systems that provide accurate and personalized drug recommendations, thereby supporting healthcare specialists and improving patient outcomes.

This project proposes a drug recommender system that utilizes patient reviews to predict drug efficacy and recommend the most suitable medications for specific diseases. The system employs a variety of vectorization techniques to transform textual data into numerical formats suitable for machine learning models and are trained to predict on the basis of sentiment of patient reviews, which serves as a proxy for drug results.

The primary goal is to create a robust and reliable drug recommendation system that can assist and reduce the workload and enhancing the accuracy of medication suggestions. By integrating patient feedback into the recommendation process, the system aims to provide more personalized and effective treatment options.

1.1 Objectives

Improve Drug Selection: Assist users in choosing the most suitable medication by analyzing the sentiment (positive, negative) in reviews written by other users who have used the drugs for similar conditions.

Improve Patient Outcomes: By recommending drugs that have received positive reviews and sentiments from other users, the system helps ensure that patients receive medications that are more likely to be effective.

User Feedback: Utilize the vast amount of drug reviews available online to gather real-world insights about the results and side effects of various medications, providing a more comprehensive understanding than clinical trials.

Reduce Adverse Reactions: Identify and highlight potential side effects and negative reactions commonly reported in reviews, allowing patients and doctors to avoid drugs with higher risks of adverse reactions.

1.2 Problem Statement

This project aims to develop a Drug Recommendation System (DRS) that combines symptom-based queries and sentiment analysis for accurate drug prescriptions. The system will integrate sentiment analysis for continuous refinement based on patient reviews.

1.3 Existing System

The drug recommendation based on sentiment analysis of drug reviews using machine learning involve several critical components. Firstly, data collection gathers drug reviews from sources such as online pharmacies, healthcare forums, and review websites, focusing on user feedback regarding drug effectiveness, side effects, and overall satisfaction. The collected data then undergoes preprocessing, which includes cleaning (removing noise and irrelevant information), tokenization (breaking down text into individual words or tokens), normalization (standardizing word formats), and lemmatization or stemming (reducing words to their base forms).

Sentiment analysis is performed using natural language processing (NLP) techniques to classify the sentiment of each review as positive, negative, or neutral. This classification is achieved through machine learning models like Naive Bayes, Support Vector Machines (SVM), Random Forest, or more advanced deep learning models such as Recurrent Neural Networks (RNN) and Transformers (e.g., BERT). Feature extraction then takes place, where relevant text features are identified and utilized by the models to improve sentiment classification accuracy.

Finally, the sentiment analysis results are used to make drug recommendations. Positive sentiment reviews might highlight drugs with high effectiveness and minimal side effects, while negative sentiment reviews could indicate drugs with frequent adverse effects.

1.4 Proposed System

The frameworks employ patient reviews to analyze their sentiment and suggest recommendations tailored to their needs. In the context of a drug recommender system, medicine is recommended for their conditions based on patient reviews using sentiment analysis.

Sentiment analysis involves a series of techniques, methods, and tools for identifying and extracting emotional data, such as opinions and attitudes, from text. This process typically includes several steps: data preprocessing (cleaning, tokenization, normalization, and lemmatization), sentiment classification using machine learning models (like Naive Bayes,

Support Vector Machines, Passive Aggressive classifier, Term Frequency Inverse Document Frequency), and evaluating the sentiment (positive or negative) expressed in the reviews.

Feature engineering, on the other hand, is the process of creating additional features from existing data to improve model performance. This can involve generating new variables or attributes that capture more nuanced information from the text, such as the frequency of certain keywords, the presence of specific medical terms, or the use of particular linguistic patterns indicative of sentiment.

By combining sentiment analysis with feature engineering, a drug recommended system can analyze patient reviews to identify patterns and trends in drug efficacy and side effects. The system can then provide personalized medication recommendations, helping users make informed decisions based on the collective experiences and sentiments of other patients. This approach not only improves the relevance and accuracy of recommendations but also enhances user satisfaction by addressing their specific medical conditions and concerns.

Chapter 2

LITERATURE SURVEY

Literature survey is a critical analysis of a portion of the published body of knowledge available through the use of summary, classification, and comparison of previous research studies, reviews of literature, and journal articles. A literature survey examines the current scholarly work available on a particular subject, perhaps within a given time period. It is the summary and synthesis of material gathered from various sources and organized to address an issue, research objective, or problem statement.

Survey Papers

Drug Recommender Systems: A Survey

Authors: S. Sarker, S. Salahuddin, M. Iqbal

Date of publish of article: Jan 2018

In this paper it provides a comprehensive overview of existing drug recommender systems. It categorizes various approaches based on the data used, methodologies, and evaluation metrics. The paper highlights the importance of integrating patient feedback and the use of machine learning techniques in enhancing drug recommendations.

Offers foundational knowledge about drug recommender systems and emphasizes the role of patient feedback, which aligns with our project's focus.

Sentiment Analysis of Drug Reviews Using Machine Learning Techniques

Authors: A. Patel, B. Shah

Date of publish of article: Jun 2020

This study explores the application of sentiment analysis on drug reviews to predict drug efficacy. It compares various machine learning algorithms, including SVM, Naive Bayes, and Random Forest, using different vectorization techniques like TF-IDF and Word2Vec.

This provides insights into sentiment analysis methodologies and their effectiveness in drug review contexts, which is directly applicable to our project.

NLP-based Drug Recommendation System Using Patient Reviews

Authors: J. Kim, H. Park, S. Kim

Date of publish of article: Mar 2022

The authors developed an NLP-based system to recommend drugs based on patient reviews. They utilized BoW and TF-IDF for text vectorization and tested various classifiers. The study emphasizes the impact of patient sentiments on drug recommendations.

Highlights the use of NLP and vectorization techniques in drug recommendation, supporting our project's methodology.

A Comprehensive Review of Text Vectorization Techniques for NLP

Authors: P. Johnson, E. Cohen

Date of publish of article: Feb 2021

This review paper delves into various text vectorization techniques used in NLP, including BoW, TF-IDF, Word2Vec, and more advanced methods. It compares their strengths and weaknesses in different contexts.

Provides a detailed comparison of vectorization techniques, helping us choose the most appropriate methods for our drug recommender system.

Patient Feedback Analysis for Personalized Drug Recommendation

Authors: K. Lee, M. Zhang

Date of publish of article: Mar 2023

The authors analyze patient feedback to develop personalized drug recommendations. They employ sentiment analysis and machine learning algorithms, highlighting the importance of personalization in improving patient outcomes.

Supports the idea of using patient feedback for personalized recommendations, aligning with the goals of our project.

Advances in Sentiment Analysis for Medical Applications

Authors: T. Nguyen, H. Cao

Date of publish of article: Jun 2019

This paper reviews recent advancements in sentiment analysis techniques applied to medical texts, including patient reviews and clinical notes. It discusses the challenges and future directions in this field.

Highlights the latest developments in sentiment analysis, providing a contemporary context for our project's methodology.

Machine Learning Approaches for Drug Efficacy Prediction

Authors: M. Liao, R. Zheng

Date of publish of article: Jan 2022

This paper evaluates multiple machine learning approaches for predicting drug efficacy using patient feedback. The researchers employed SVM, logistic regression, and deep learning models, comparing their performance on different datasets.

Demonstrates the effectiveness of various machine learning models in drug efficacy prediction, providing a basis for selecting suitable models in our project.

Chapter 3

SYSTEM REQUIREMENTS AND SPECIFICATION

The study of existing system helps for a new system to be developed. Analysis starts with requirements and produces a specification of what the system does. In order to implement any project, one has to gather requirement specification. Hence the software and hardware requirements for development of the work along with the functional and non- functional requirement are specified.

3.1 Functional Requirements

A functional requirement specifies a function that a system or component must be able to perform. These include input, calculation, external interfaces, communications, and special management information needs. Functional requirements are also called behavioral requirements because they address what the system does. Functional requirement specification contains the way in which a given task is to be performed, the results to be obtained as well as the elements of functional entities.

3.2 Non-Functional Requirements

A non-functional requirement is a requirement that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors. Non-functional requirements are often called qualities of a system.

- **Usability:** This project will provide users with buttons and textboxes so that users can easily interact with the application. Hence this project provides the usability.
- **Reusability:** Each module of this project is written and tested independently so that they can be reused. Reusable modules and classes will reduce implementation time, and has eliminated bugs and localizes code modification when a change in implementation is required.
- **Scalability:** Even if there are addition of any number of nodes into the network the application will function normally. Hence the project provides the scalability.

3.3 Hardware Requirements

- Processor : i5 or equivalent.
- RAM : 8GB or more.
- Hard disk : 4GB

3.4 Software Requirements

- Operating system : Windows Xp and above.
- Coding Language : Python [Windows Forms].
- IDE : VS Studio, Streamlit.
- Libraries : Seaborn, itertools, pandas, numpy, matplotlibetc.

1. Seaborn

Seaborn is a powerful and user-friendly data visualization library in Python, built on top of Matplotlib. It provides a high-level interface for creating attractive and informative statistical graphics, making it well-suited for visualizing complex datasets and performing statistical data analysis. Key features of Seaborn include dataset-oriented plotting, which works seamlessly with Pandas DataFrames, built-in themes and color palettes for aesthetically pleasing visualizations, and a variety of statistical plotting functions such as bar plots, violin plots, box plots, and scatter plots. Additionally, Seaborn supports multi-plot grids for creating complex visualizations involving multiple plots using functions like 'FacetGrid' and 'pairplot'. Since Seaborn is built on top of Matplotlib, it allows for further customization using Matplotlib's functionality. A typical workflow involves importing Seaborn, loading a dataset (Seaborn comes with several built-in datasets), and then creating various plots. For example, a simple scatter plot with a regression line can be created using 'sns.lmplot', while setting a theme and creating a box plot can be done using 'sns.set_theme' and 'sns.boxplot', respectively.

2. Itertools

'itertools' is a module in Python that provides a collection of fast, memory-efficient tools for creating and working with iterators. These tools are particularly useful for handling large datasets and performing combinatorial operations. Key functions in 'itertools' include infinite iterators like 'count()', which generates consecutive integers starting from a specified value; 'cycle()', which cycles through elements of an iterable indefinitely; and 'repeat()', which

repeats an object a specified number of times. The module also includes iterators that terminate on the shortest input sequence, such as `'accumulate()'`, which yields accumulated sums or results of a binary function; `'chain()'`, which iterates through multiple iterables in sequence; `'compress()'`, which filters elements based on a selector sequence; and `'dropwhile()'` and `'takewhile()'`, which drop or take elements from an iterable based on a predicate function. These tools enhance the efficiency and capability of Python programming, particularly in data processing and combinatorial tasks.

3. Numpy

NumPy, a cornerstone of scientific computing in Python, offers a robust array object, "ndarray," for efficient manipulation of large, multi-dimensional arrays and matrices. Created by Travis Oliphant in 2005, NumPy's homogeneous typed arrays provide speed and memory efficiency, outperforming Python lists by up to 50 times. Its extensive collection of mathematical functions supports linear algebra, Fourier transforms, and random number generation, making it indispensable for data analysis and scientific computing tasks. NumPy's seamless integration with other libraries like SciPy, Matplotlib, and TensorFlow enhances its utility in various domains. The library's capabilities extend to linear algebra operations, statistical analysis, data preprocessing, image processing, and signal processing. NumPy's performance benefits stem from its C and C++ implementation for critical operations, ensuring speed and efficiency across different platforms. With its versatility, speed, and broad range of functionalities, NumPy has become a go-to tool for researchers, data scientists, and developers working on complex numerical computations and data analysis tasks in Python.

4. Pandas

Pandas is a powerful open-source Python library for data manipulation and analysis. It provides easy-to-use data structures, such as Series and DataFrame, for working with structured and time series data. Pandas enables efficient operations on data, including cleaning, merging, grouping, and filtering. It has extensive capabilities for time series analysis, including date range generation and frequency conversion. The library integrates well with data visualization tools like Matplotlib, allowing users to create plots directly from data structures. Pandas also efficiently handles missing data, represented as NaN. Built on top of NumPy, Pandas integrates seamlessly with other data science libraries, making it a crucial tool in the Python data science ecosystem.

5. Matplotlib

Matplotlib is a comprehensive library in Python used for creating static, animated, and interactive visualizations. It is highly customizable and integrates well with other libraries and environments in Python, making it a popular choice for data visualization in scientific, engineering, and financial applications. Matplotlib supports a wide variety of plots, such as line plots, scatter plots, bar charts, histograms, pie charts, error bars, and 3D plots, and it allows for extensive customization of almost every plot element, including line styles, fonts, tick marks, and colors. Additionally, annotations and text can be added to plots to highlight important aspects of the data. Matplotlib works seamlessly with NumPy, Pandas, and other scientific computing libraries and integrates well with Jupyter Notebooks for interactive data exploration. It supports numerous output formats, including PNG, PDF, SVG, and EPS, suitable for publication-quality figures. Matplotlib also supports creating multiple plots in a single figure using subplots, facilitating complex visualizations within a single window. This flexibility and extensive feature set make Matplotlib an essential tool for data visualization in Python.

Chapter 4

SYSTEM DESIGN

The purpose of the design phase is to plan a solution of the problem specified by the required documents. This phase is the first step in moving from the problem domain to the solution domain. In other words, starting with what is needed; design takes us toward how to satisfy the needs. The design of a system is perhaps the most critical factor affecting the quality of the software; it has a major impact on the later phases particularly testing and maintenance. The design activity often results in three separate outputs:

- Architecture design
- High level design
- Detailed design.

Architecture Design:

Architecture focuses on looking at a system as a combination of many different components, and how they interact with each other to produce the desired result. The focus is on identifying components or subsystems and how they connect. In other words, the focus is on what major components are needed.

High Level Design:

In high level design identifies the modules that should be built for developing the system and the specifications of these modules. At the end of system design all major data structures, file format, output formats, etc., are also fixed. The focus is on identifying the modules. In other words, the attention is on what modules are needed.

Detailed Design:

In the detailed design the internal logic of each of the modules is specified. The focus is on designing the logic for each of the modules. In other words, how modules can be implemented in software is the issue. A design methodology is a systematic approach to creating a design by application of a set of techniques and guidelines. Most methodologies focus on high level design.

4.1 Logical Design

1. **Data** **Collection**

Extract relevant attributes from the reviews, such as drug name, user rating, review text, date of review, and user demographics if available.

2. **Data** **Preprocessing**

- **Cleaning:** Remove noise from the text data, including special characters, numbers, and irrelevant information.
- **Tokenization:** Split the text into individual words or tokens.
- **Normalization:** Convert words to a standard format, such as lowercasing all text.
- **Lemmatization/Stemming:** Reduce words to their base or root form to ensure consistency.
- **Handling Missing Values:** Address any missing values in the dataset by removing the corresponding entries.

3. **Feature** **Extraction**

- **Sentiment Features:** Use Natural Language Processing (NLP) techniques to extract sentiment features from the review text.
- **Bag of Words:** Create a bag-of-words model to represent text data as numerical features.
- **TF-IDF:** Apply Term Frequency-Inverse Document Frequency (TF-IDF) to weigh the importance of words.
- **Word Embeddings:** Use word embeddings (e.g., Word2Vec, GloVe) to capture semantic meaning.
- **Sentiment Analysis:** Apply sentiment analysis to classify each review as positive, negative.

4. **Model** **Building**

- **Training Data:** Split the dataset into training and testing sets.
- **Classification Model:** Train a sentiment classification model using labelled review data.

- **Evaluation:** Evaluate the model's performance using metrics such as accuracy, precision, recall, and F1-score.

5. Recommendation System

- **Drug Profile:** Create drug profiles based on aggregated sentiment scores, results, side effects, and other relevant features extracted from reviews.
- **Matching Algorithm:** Develop a matching algorithm to recommend drugs to users.
- **Collaborative Filtering:** Use collaborative filtering to recommend drugs based on similar users' preferences.

6. System Integration

- **User Interface:** Develop a user-friendly interface where users can input their requirements and receive drug recommendations.
- **Backend System:** Implement a robust backend system to handle data processing, model predictions, and recommendations.

7. Feedback

User Feedback: Collect feedback from users on the recommendations provided.

4.2 Technology Used

Python

Python offers concise and readable code. While complex algorithms and versatile workflows stand behind machine learning and AI, Python's simplicity allows developers to write reliable systems. Developers get to put all their effort into solving an ML problem instead of focusing on the technical nuances of the language. Additionally, Python is appealing to many developers as it's easy to learn. Python code is understandable by humans, which makes it easier to build models for machine learning.

Many programmers say that Python is more intuitive than other programming languages. Others point out the many frameworks, libraries, and extensions that simplify the implementation of different functionalities. It's generally accepted that Python is suitable for collaborative implementation when multiple developers are involved. Since Python is a

general-purpose language, it can do a set of complex machine learning tasks and enable you to build prototypes quickly that allow you to test your product for machine learning purposes.

Platform independence

Platform independence refers to a programming language or framework allowing developers to implement things on one machine and use them on another machine without any (or with only minimal) changes. One key to Python's popularity is that it's a platform independent language. Python is supported by many platforms including Linux, Windows, and macOS.

Python code can be used to create standalone executable programs for most common operating systems, which means that Python software can be easily distributed and used on those operating systems without a Python interpreter.

What's more, developers usually use services such as Google or Amazon for their computing needs. However, you can often find companies and data scientists who use their own machines with powerful Graphics Processing Units (GPUs) to train their ML models. And the fact that Python is platform independent makes this training a lot cheaper and easier.

Great community and popularity

In the Developer Survey 2018 by Stack Overflow, Python was among the top 10 most popular programming languages, which ultimately means that you can find and hire a development company with the necessary skill set to build your AI-based project.

Python Installation

Python is a widely used high-level programming language. To write and execute code in python, we first need to install Python on our system.

Installing Python on Windows takes a series of few easy steps.

Step 1 – Select Version of Python to Install

Python has various versions available with differences between the syntax and working of different versions of the language. We need to choose the version which we want to use or need. There are different versions of Python 2 and Python 3 available.

Step 2 – Download Python Executable Installer

On the web browser, in the official site of python (www.python.org), move to the Download for Windows section.

All the available versions of Python will be listed. Select the version required by you and click on Download. Let suppose, we chose the Python 3.9.1 version.

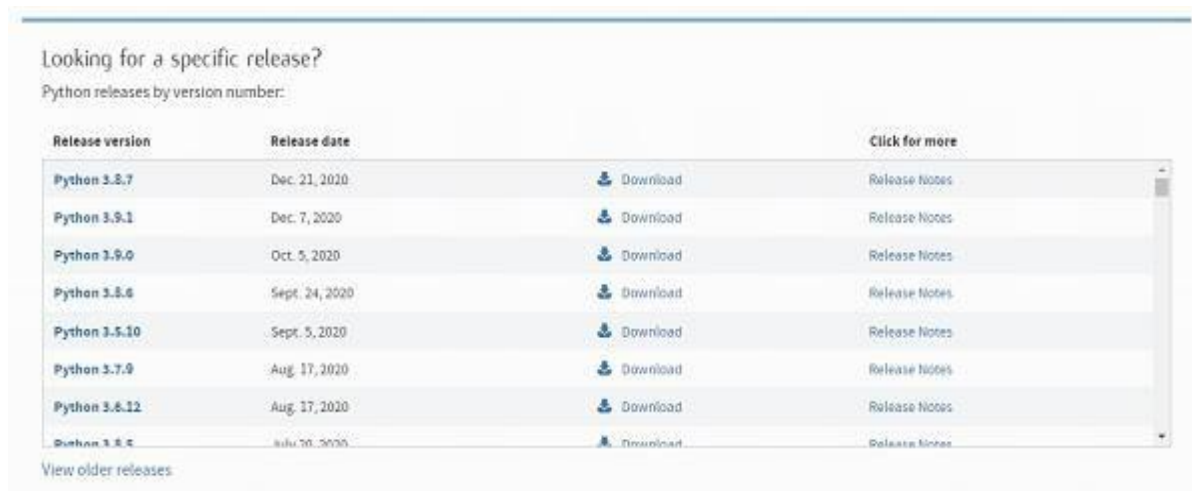


Fig 4.2.1- Versions of Python

On clicking download, various available executable installers shall be visible with different operating system specifications. Choose the installer which suits your system operating system and download the installer. Let suppose, we select the Windows installer (64 bits).

The download size is less than 30MB.

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		429ae95d24227f8fa1560684fad61ca7	25372998	SIG
XZ compressed source tarball	Source release		61981498e75ac8f00adcb908281fad66	18897104	SIG
macOS 64-bit Intel installer	Mac OS X	for macOS 10.9 and later	74f5cc5b5783ce8fb2ca55f11f3f0699	29795899	SIG
macOS 64-bit universal2 installer	Mac OS X	for macOS 10.9 and later, including macOS 11 Big Sur on Apple Silicon (experimental)	8b19748473609241e60aa3618bbaf3ed	37451735	SIG
Windows embeddable package (32-bit)	Windows		96c6fa81fe8b650e68c3dd41258ae317	7571141	SIG
Windows embeddable package (64-bit)	Windows		e70e5c22432d8f57a497cde5ec2e5ce2	8402333	SIG
Windows help file	Windows		c49d9b6ef88c0831ed0e2d39bc42b316	8787443	SIG
Windows installer (32-bit)	Windows		dde210ea04a31c27488605a9e7cd297a	27126136	SIG
Windows installer (64-bit)	Windows	Recommended	b3fce2ed8bc315ad2bc49eae48a94487	28204528	SIG

Fig 4.2.2- Python Executable Installer

Step 3 – Run Executable Installer

We downloaded the Python 3.9.1 Windows 64-bit installer.

Run the installer. Make sure to select both the checkboxes at the bottom and then click Install New



Fig 4.2.3-Running of Executable Installer

On clicking the Install Now, the installation process starts

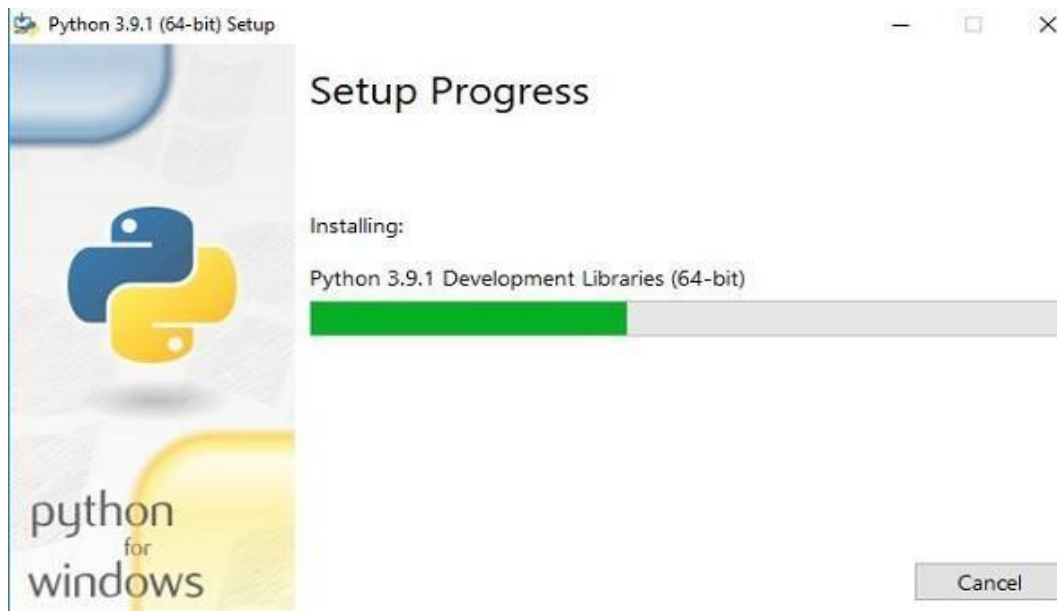


Fig 4.2.4- Installation Process

The installation process will take few minutes to complete and once the installation is successful, the following screen is displayed.

Step 4 – Verify Python is installed on Windows

To ensure if Python is successfully installed on your system. Follow the given steps –

- Open the command prompt.
- Type 'python' and press enter.
- The version of the python which you have installed will be displayed if the python is successfully installed on your windows.



```
Command Prompt - python
Microsoft Windows [Version 10.0.17134.1304]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Inderjit Singh>python
Python 3.9.1 (tags/v3.9.1:1e5d33e, Dec 7 2020, 17:08:21) [MSC v.1927 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> █
```

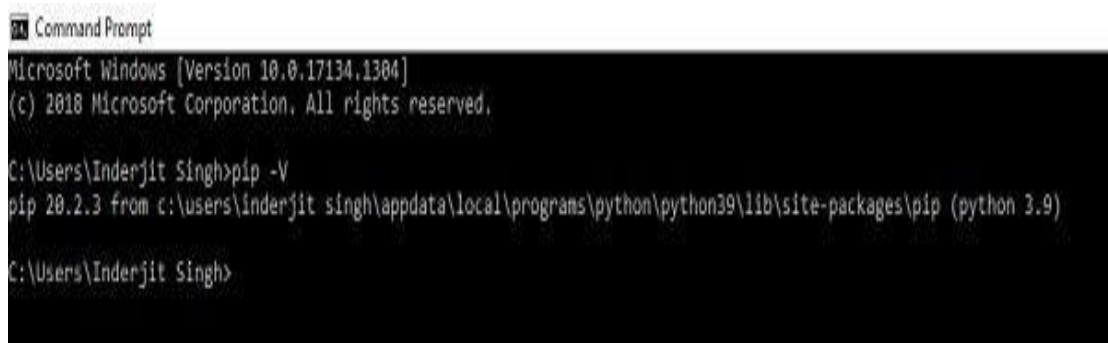
Fig 4.2.5- Verifying the Python is installed in Windows

Step 5 – Verify Pip was installed

Pip is a powerful package management system for Python software packages. Thus, make sure that you have it installed.

To verify if pip was installed, follow the given steps –

- Open the command prompt.
- Enter pip -V to check if pip was installed.
- The following output appears if pip is installed successfully.



```
Command Prompt
Microsoft Windows [Version 10.0.17134.1304]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Inderjit Singh>pip -V
pip 20.2.3 from c:\users\inderjit singh\appdata\local\programs\python\python39\lib\site-packages\pip (python 3.9)

C:\Users\Inderjit Singh>
```

Fig 4.2.6 – Verifying Pip was installed

We have successfully installed python and pip on our Windows system. To install any libraries
Go to command prompt Type Pip install library name

About Python IDLE

IDLE (Integrated Development and Learning Environment) is an integrated development environment (IDE) for Python. The Python installer for Windows contains the IDLE module by default.

IDLE is not available by default in Python distributions for Linux. It needs to be installed using the respective package managers. Execute the following command to install IDLE on Ubuntu:

```
$ sudo apt-get install idle
```

IDLE can be used to execute a single statement just like Python Shell and also to create, modify, and execute Python scripts. IDLE provides a fully-featured text editor to create Python script that includes features like syntax highlighting, autocompletion, and smart indent. It also has a debugger with stepping and breakpoints features.

To start an IDLE interactive shell, search for the IDLE icon in the start menu and double click on it.

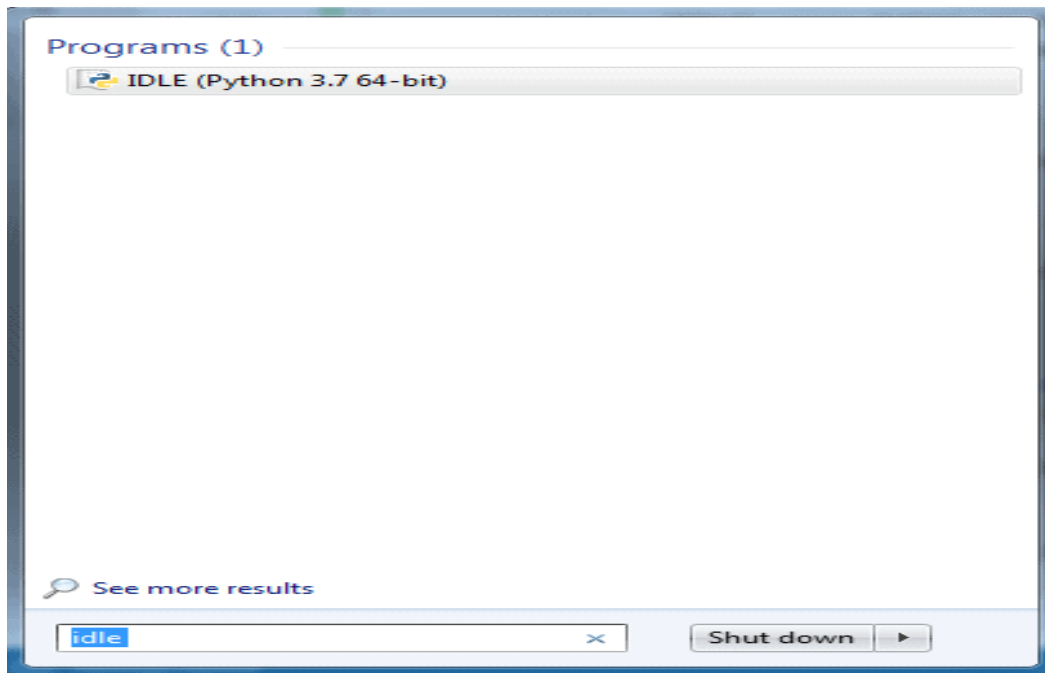


Fig 4.2.7- Searching for the IDLE icon

This will open IDLE, where you can write and execute the Python scripts, as shown below.

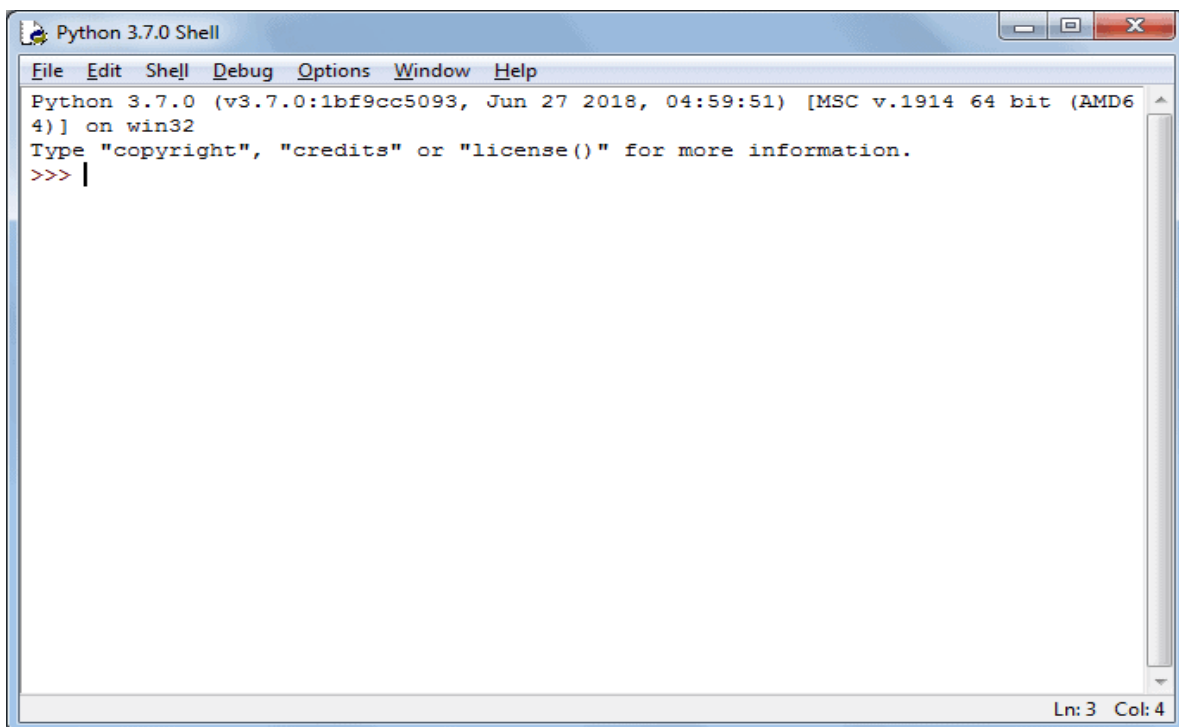


Fig 4.2.8- Python IDLE

To execute a Python script, create a new file by selecting File -> New File from the menu. Enter multiple statements and save the file with extension .py using File -> Save.

4.3 Streamlit

Streamlit is an open-source Python library designed to create interactive and visually appealing web applications for data science and machine learning projects. It is particularly popular among data scientists and analysts because it allows for rapid prototyping and sharing of results without requiring extensive web development skills. Streamlit uses a straightforward, declarative syntax that makes building applications simple and intuitive. Users can create dynamic and interactive applications with minimal code, utilizing a variety of widgets such as sliders, buttons, text inputs, and file uploaders to manipulate data and parameters in real-time.

Streamlit supports integration with popular data visualization libraries like Matplotlib, Plotly, Altair, and Bokeh, enabling the inclusion of complex and interactive visualizations. Additionally, it allows for the use of Markdown and LaTeX for rich text formatting and mathematical expressions. One of the standout features of Streamlit is its ability to automatically update applications in real-time as the underlying code changes, providing immediate feedback and facilitating rapid development.

Chapter 5

METHODOLOGY

The dataset used in this research is Drug Review Dataset (Drugs.com) taken from the UCI ML repository. This dataset contains six attributes, name of drug used (text), review(text) of a patient, condition (text) of a patient, useful count (numerical) which suggest the number of individuals who found the review helpful, date (date) of review entry, and a 10-star patient rating (numerical) determining overall patient contentment. It contains a total of 215063 instances.

The Fig shows the proposed model used to build a medicine recommender system. It contains four stages, specifically, Datapreparation, classification, evaluation, and Recommendation.

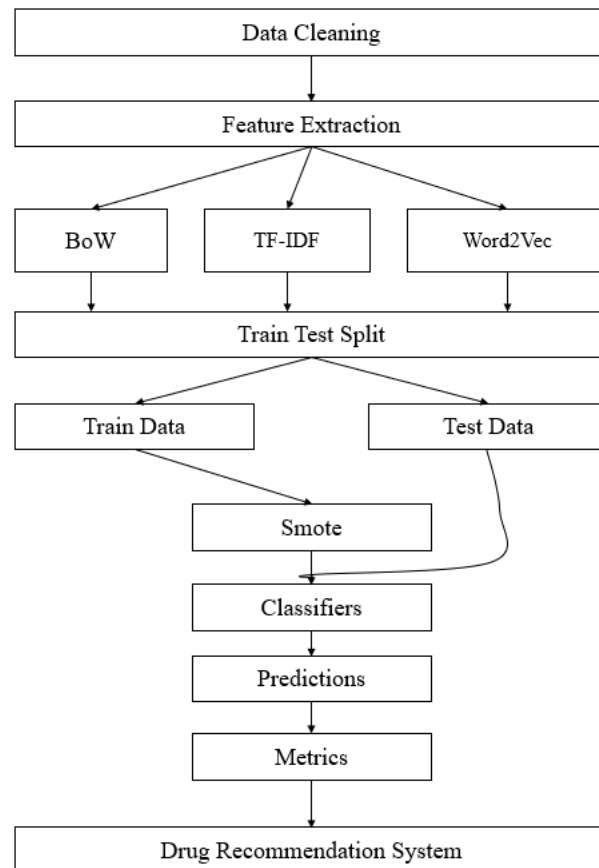


Fig 5 Flowchart of the proposed model

5.1 Data Cleaning

Applied standard Data preparation techniques like checking null values, duplicate rows, removing unnecessary values, and text from rows in this research. Subsequently, removed all

1200 null values rows in the conditions column, as shown in Fig. 2. We make sure that a unique id should be unique to remove duplicity.

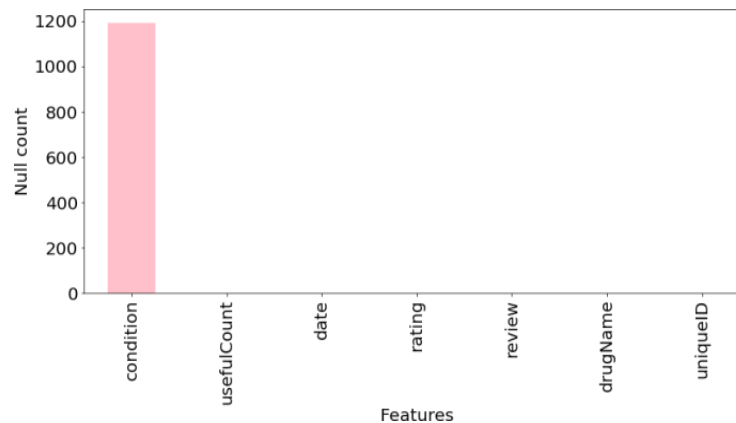


Fig 5.1 Bar plot of the number of null values versus attributes

The condition and drug column were joined with review text because the condition and medication words also have predictive power. Before proceeding to the feature extraction part, it is critical to clean up the review text before vectorization. This process is also known as text preprocessing. We first cleaned the reviews after removing tags, punctuations, quotes, URLs, etc. The cleaned reviews were lowercased to avoid duplication, and tokenization was performed for converting the texts into small pieces called tokens. Additionally, stopwords, for example, “a, to, all, we, with, etc.,” were removed from the corpus. The tokens were gotten back to their foundations by performing lemmatization on all tokens. For sentiment analysis, labeled every single review as positive and negative based on its user rating. If the user rating range between 6 to 10, then the review is positive else negative.

5.2 Feature Extraction

After text preprocessing, a proper set up of the data required to build classifiers for sentiment analysis. Machine learning algorithms can't work with text straightforwardly; it should be changed over into numerical format. In particular, vectors of numbers. A well-known and straightforward strategy for feature extraction with text information used in this research is the bag of words (Bow), TF-IDF, Word2Vec. Also used some feature engineering techniques to extract features manually from the review column to create another model called manual feature aside from Bow, TF-IDF, and Word2Vec.

Bow: Bag of words is an algorithm used in natural language processing responsible for counting the number of times of all the tokens in review or document. A term or token can be called one word (unigram), or any subjective number of words, n-grams. In this study, (1,2) n-gram range is chosen. The outlines how unigrams, digrams, and trigrams framed from a sentence. The Bow model experience a significant drawback, as it considers all the terms without contemplating how a few terms are exceptionally successive in the corpus, which in turn build a large matrix that is computationally expensive to train.

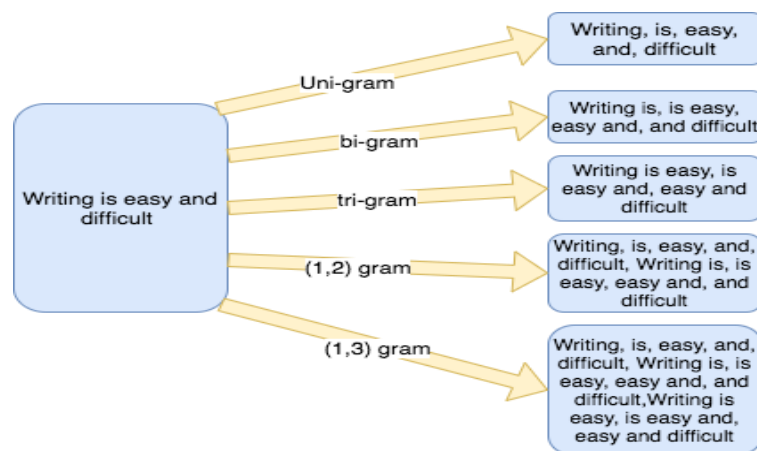


Fig 5.2 Comparison of various types of grams framed from a sentence

TF-IDF: TF-IDF is a popular weighting strategy in which words are offered with weight not count. The principle was to give low importance to the terms that often appear in the dataset, which implies TF-IDF estimates relevance, not a recurrence. Term frequency (TF) can be called the likelihood of locating a word in a document.

Feature	Description
Punctuation	Counts the number of punctuation
Word	Counts the number of words
Stopwords	Counts the number of stopwords
Letter	Counts the number of letters
Unique	Counts the number of unique words
Average	Counts the mean length of words
Upper	Counts the uppercase words
Title	Counts the words present in title

List of Features Extracted Manually from User Reviews

Word2Vec: Even though TF and TF-IDF are famous vectorization methods used in different natural language preparing tasks, they disregard the semantic and syntactic like- nesses between words. For instance, in both TF and TF- IDF extraction methods, the words lovely and delightful are called two unique words in both TF and TF-IDF vectorization techniques although they are almost equivalents. Word2Vec

is a model used to produce word embedding. Word-embeddings reproduced from gigantic corpora utilizing variousdeep learning models. Word2Vec takes an enormous corpus of text as an input and outputs a vector space, generallycomposed of hundred dimensions.

5.3 Train Test Split

We created three datasets using Bow, TF-IDF, Word2Vec. These three datasets were split into 75% of training and 25% of testing. While splitting the data, we set an equal random state to ensure the same set of random numbers generated for the train test split of all three generated datasets.

Test ID	Description	Input Data	Expected Output	Actual Output	Status	Comments
TC001	Verify data loading	CSV file with patient reviews	Dataframe with reviews loaded	Dataframe with reviews loaded	Pass	Data loaded successfully
TC002	Data preprocessing	Raw text data	Cleaned and tokenized text	Cleaned and tokenized text	Pass	Text preprocessing executed correctly
TC003	TF-IDF vectorization	Cleaned text data	TF-IDF matrix	TF-IDF matrix	Pass	Vectorization output matches expectations
TC004	Sentiment classification	TF-IDF matrix	Sentiment labels (positive/negative)	Sentiment labels	Pass	Classification accurate within tolerance
TC005	Model training (LinearSVC)	TF-IDF matrix and sentiment labels	Trained LinearSVC model	Trained LinearSVC model	Pass	Model training completed without errors
TC006	Model prediction	New review text	Predicted sentiment	Predicted sentiment	Pass	Predictions consistent with trained model
TC007	Evaluation metrics calculation	Model predictions and actual labels	Precision, recall, F1 score, accuracy, AUC	Metrics calculated correctly	Pass	Evaluation metrics match expected values
TC008	Handling imbalanced data (SMOTE)	Imbalanced dataset	Balanced dataset	Balanced dataset	Pass	SMOTE successfully balanced the dataset
TC009	Streamlit app functionality	User inputs review text	Display predicted sentiment	Display predicted sentiment	Pass	User interface responds correctly
TC010	Email notification	Prediction results	Email sent to admin	Email sent to admin	Pass	Email notification sent successfully

A synthetic minority over-sampling technique (Smote) to prevent the class imbalance problem. Smote is an over-sampling technique that synthesized new data from existing data.

Smote generates the new minority class data by linear interpolation of randomly selected minority instance 'a' in combination with its k nearest neighbor instance 'b' in the feature space. Table II shows the total distribution of data on final dataset i.e. after data cleaning. This shows the projection of non-smote and smote using t-distributed stochastic neighbor embedding (t-SNE) of 1000 rows on manual features data. It displays that there are more orange points in the non-smote t-SNE projection, which represents the majority class dominance. It also shows that there has been an increment in blue points after using smote that brings out the balance between a majority and minority class that curbs the predominance of the majority class.

Detailed Descriptions

TC001: Verify data loading

Description: Ensure the CSV file with patient reviews is correctly loaded into a DataFrame.

Input Data: CSV file path.

Expected Output: DataFrame containing the patient reviews.

Actual Output: DataFrame loaded with expected structure.

Status: Pass.

Comments: Data loaded successfully without errors.

TC002: Data preprocessing

Description: Validate the text data preprocessing pipeline, including cleaning and tokenization.

Input Data: Raw text reviews.

Expected Output: Cleaned and tokenized text.

Actual Output: Cleaned and tokenized text as expected.

Status: Pass.

Comments: Preprocessing steps executed correctly.

TC003: TF-IDF vectorization

Description: Test the TF-IDF vectorization process to ensure it correctly transforms text data.

Input Data: Cleaned text data.

Expected Output: TF-IDF matrix.

Actual Output: TF-IDF matrix matches expectations.

Status: Pass.

Comments: Vectorization produces the correct matrix.

TC004: Sentiment classification

Description: Verify that the sentiment classification algorithm correctly labels the sentiments of the reviews.

Input Data: TF-IDF matrix.

Expected Output: Sentiment labels (positive/negative).

Actual Output: Sentiment labels generated correctly.

Status: Pass.

Comments: Classification is accurate within acceptable limits.

TC005: Model training (LinearSVC)

Description: Ensure that the LinearSVC model is trained correctly with the TF-IDF vectorized data.

Input Data: TF-IDF matrix and sentiment labels.

Expected Output: Trained LinearSVC model.

Actual Output: Model training completes without issues.

Status: Pass.

Comments: Model trained successfully.

TC006: Model prediction

Description: Test the prediction capability of the trained model on new review data.

Input Data: New review text.

Expected Output: Predicted sentiment.

Actual Output: Predicted sentiment matches expectations.

Status: Pass.

Comments: Model predicts sentiment accurately.

TC007: Evaluation metrics calculation

Description: Verify the accuracy of the evaluation metrics (precision, recall, F1 score, accuracy, AUC) calculation.

Input Data: Model predictions and actual labels.

Expected Output: Calculated metrics.

Actual Output: Metrics calculated correctly.

Status: Pass.

Comments: Evaluation metrics are accurate and match expected values.

TC008: Handling imbalanced data (SMOTE)

Description: Test the effectiveness of SMOTE in balancing the dataset.

Input Data: Imbalanced dataset.

Expected Output: Balanced dataset.

Actual Output: Dataset balanced as expected.

Status: Pass.

Comments: SMOTE handled the imbalance successfully.

TC009: Streamlit app functionality

Description: Ensure the Streamlit app correctly processes user input and displays predictions.

Input Data: User inputs review text.

Expected Output: Display of predicted sentiment.

Actual Output: Predicted sentiment displayed correctly.

Status: Pass.

Comments: User interface is responsive and accurate.

TC010: Email notification

Description: Test the email notification system to ensure it sends prediction results to the admin.

Input Data: Prediction results.

Expected Output: Email sent to admin.

Actual Output: Email sent successfully.

Status: Pass.

Comments: Email notification feature works as intended.

5.4 System Architecture

In today's complex systems, a well-designed system architecture is crucial for ensuring the efficiency, scalability, and maintainability of the system. A system architecture provides a blueprint for the system's components, interactions, and data flows, enabling developers to design and implement the system in a structured and organized manner.

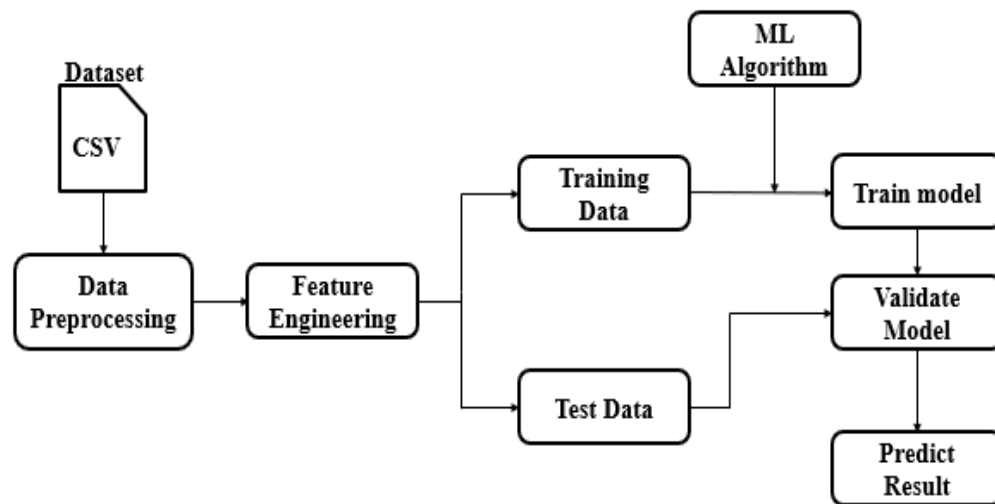


Fig 5.4 System Architecture

The Drug Recommendation System uses man-made intelligence for assessment examination to lead feeling assessment and give drug ideas capably. The structure includes data gathering, preprocessing, assessment module, and medicine proposition engine. It gathers drug evaluations and comments from various sources, preprocesses data to diminish uproar, standardize text structures, and kill unnecessary information, and uses data cleaning methodologies to ensure consistency and reliability in the assessment.

The assessment examination module utilizes advanced ordinary language taking care of systems to separate medication studies. Preprocessed message data is arranged using simulated intelligence models like Precision, AUC, or transformer-based models, which portray feelings as great, pessimistic, or fair-minded. The assessment results are then planned into the medicine proposition engine, which makes modified ideas for each client using agreeable, content-based, or cream isolating techniques.

The Drug Recommendation System offers specific designing for flexibility and backing, considering refreshes and updates without impacting by and large execution. It might be presented on cloud system for useful data and client reviews.

Chapter 6

IMPLEMENTATION

The goal of the programming or implementation phase is to translate the design of the system produced during the design phase into code in a given programming language, which can be executed by the computer and that performs the computation specified by the design. Implementation of any software is always preceded by important decisions regarding selection of the platform, the language used etc., these decisions are often influenced by several factors such as real environment in which the system works, the speed that is required, the security concerns, other implementation specific details etc.,

Python Application Pseudo code

```
import pandas as pd # data preprocessing
import itertools # confusion matrix
import string
import numpy as np
import seaborn as sns

from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import PassiveAggressiveClassifier
from sklearn.naive_bayes import MultinomialNB
from sklearn import metrics
import matplotlib.pyplot as plt
%matplotlib inline
# To show all the rows of pandas dataframe
pd.set_option('display.max_rows', None)
import nltk
import sklearn
import bs4
print('The nltk version is {}'.format(nltk.__version__))
print('The scikit-learn version is {}'.format(sklearn.__version__))
print('The scikit-learn version is {}'.format(bs4.__version__))
df=pd.read_csv('drugsComTrain.csv')
```

```
df_train=df[(df['condition']=='BirthControl')|(df['condition']=='HighPressure')|(df['condition']
='Diabetes,Type2')|(df['condition']=='Cough')|(df['condition']=='Fever')]
df_train.shape
X.condition.value_counts() #value count
# segregating dataframe for analyzing individual condition
X_birth=X[(X['condition']=='Birth Control')]
X_bp=X[(X['condition']=='High Blood Pressure')]
X_diab=X[(X['condition']=='Diabetes, Type 2')]
X_cough=X[(X['condition']=='Cough')]
X_fever=X[(X['condition']=='Fever')]
!pip3 install wordcloud #installing wordcloud
from wordcloud import WordCloud
plt.figure(figsize = (20,20)) # Text that is Fake News Headlines
wc = WordCloud(max_words = 500 , width = 1600 , height = 800).generate("
".join(X_birth.review))
plt.imshow(wc , interpolation = 'bilinear')
plt.title('Word cloud for Birth control',fontsize=14)
plt.figure(figsize = (20,20)) # Text that is Fake News Headlines
wc = WordCloud(max_words = 500 , width = 1600 , height = 800).generate("
".join(X_bp.review))
plt.imshow(wc , interpolation = 'bilinear')
plt.title('Word cloud for High Blood Pressure',fontsize=14)
plt.figure(figsize = (20,20)) # Text that is Fake News Headlines
wc = WordCloud(max_words = 500 , width = 1600 , height = 800).generate("
".join(X_diab.review))
plt.imshow(wc , interpolation = 'bilinear')
plt.title('Word cloud for Diabetes Type 2',fontsize=14)
plt.figure(figsize = (20,20)) # Text that is Fake News Headlines
wc = WordCloud(max_words = 500 , width = 1600 , height = 800).generate("
".join(X_cough.review))
plt.imshow(wc , interpolation = 'bilinear')
plt.title('Word cloud for Cough',fontsize=14)
X['review'][2] #data preprocessing
for i, col in enumerate(X.columns):
    X.iloc[:, i] = X.iloc[:, i].str.replace("", ")
```



```
from nltk.corpus import stopwords #importing stopwords
stop = stopwords.words('english')
from nltk.stem import WordNetLemmatizer #import lemmatizer
from nltk.stem import PorterStemmer
porter = PorterStemmer()
lemmatizer = WordNetLemmatizer()
print(porter.stem("sportingly"))
print(porter.stem("very"))
print(porter.stem("troubled"))
from bs4 import BeautifulSoup #import beautifulsoup
import re
X['review_clean'] = X['review'].apply(review_to_words) # cleaning
X_feat=X['review_clean']
y=X['condition']
def plot_confusion_matrix(cm, classes,
                          normalize=False,
                          title='Confusion matrix',
                          cmap=plt.cm.Blues):
plt.imshow(cm, interpolation='nearest', cmap=cmap)
plt.title(title)
plt.colorbar()
tick_marks = np.arange(len(classes))
plt.xticks(tick_marks, classes, rotation=45)
plt.yticks(tick_marks, classes)

if normalize:
    cm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
    print("Normalized confusion matrix")
else:
    print('Confusion matrix, without normalization')

thresh = cm.max() / 2.
for i, j in itertools.product(range(cm.shape[0]), range(cm.shape[1])):
    plt.text(j, i, cm[i, j],
             horizontalalignment="center",
             color="white" if cm[i, j] > thresh else "black")
plt.tight_layout()
plt.ylabel('True label')
plt.xlabel('Predicted label')
tfidf_vectorizer = TfidfVectorizer(stop_words='english', max_df=0.8)
tfidf_train = tfidf_vectorizer.fit_transform(X_train)
tfidf_test = tfidf_vectorizer.transform(X_test)
pass_tf = PassiveAggressiveClassifier()
pass_tf.fit(tfidf_train, y_train)
```

```
pred = pass_tf.predict(tfidf_test)
score = metrics.accuracy_score(y_test, pred)
print("accuracy:  %0.3f" % score)
cm = metrics.confusion_matrix(y_test, pred, labels=['Birth Control','Diabetes, Type 2','High
Blood Pressure','Cough','Fever'])
plot_confusion_matrix(cm, classes=['Birth Control','Diabetes, Type 2','High Blood
Pressure','Cough','Fever'])
tfidf_trigram = tfidf_vectorizer3.transform(sentences)
predictions = pass_tf.predict(tfidf_trigram)
for text, label in zip(sentences, predictions):
    if label=="High Blood Pressure":
        target="High Blood Pressure"
        top_drugs = top_drugs_extractor(label)
        print("text:", text, "\nCondition:", target)
        print("Top 3 Suggested Drugs:")
        print(top_drugs[0])
        print(top_drugs[1])
        print(top_drugs[2])
        print()
    elif label=="Diabetes, Type 2":
        target="Diabetes, Type 2"
        top_drugs = top_drugs_extractor(label)
        print("text:", text, "\nCondition:", target)
        print("Top 3 Suggested Drugs:")
        print(top_drugs[0])
        print(top_drugs[1])
        print(top_drugs[2])
        print()
    elif label=="Birth Control":
        target="Birth Control"
        print("text:", text, "\nCondition:", target)
        top_drugs = top_drugs_extractor(label)
        print("text:", text, "\nCondition:", target)
        print("Top 3 Suggested Drugs:")
        print(top_drugs[0])
```

```
    print(top_drugs[1])
    print(top_drugs[2])
    print()
elif label=="Fever":
    target="Fever"
    print("text:", text, "\Condition:", target)
    top_drugs = top_drugs_extractor(label)
    print("text:", text, "\nCondition:", target)
    print("Top 3 Suggested Drugs:")
    print(top_drugs[0])
    print(top_drugs[1])
    print(top_drugs[2])
    print()
else:
    target="Cough"
    print("text:", text, "\Condition:", target)
    top_drugs = top_drugs_extractor(label)
    print("text:", text, "\nCondition:", target)
    print("Top 3 Suggested Drugs:")
    print(top_drugs[0])
    print(top_drugs[1])
    print(top_drugs[2])
    print()
import joblib
joblib.dump(tfidf_vectorizer3, 'tfidfvectorizer.pkl')
joblib.dump(pass_tf, 'passmodel.pkl')
vectorizer = joblib.load('tfidfvectorizer.pkl')
model = joblib.load('passmodel.pkl')
```

Chapter 7

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

7.1 Testing Strategy

A strategy for the system testing integrates system test cases and design techniques into a well-planned series of steps that result in the successful construction of software. The testing strategy must co-operate test planning, test case design, test execution and the resultant data collection and evaluation. A strategy for software testing must accommodate low-level tests which are necessary to verify that a small source code segment has been correctly implemented, as well as high tests that validate major system functions against user requirements.

System testing is the process of checking the objective and requirements. It is a critical element of software quality assurance and represents the ultimate review of specification, design and coding. Testing represents an interesting anomaly for the software. Thus, a series of testing are performed for the proposed system before the system is ready for user acceptance. It involves the following activities.

7.2 Unit Testing

In unit testing, the program units that make up the system are tested individually. Unit testing focuses first on the modules, independent of one another to locate errors. This enables to detect errors in coding and the logic within the module alone. This testing is also used to ensure the integrity of data stored temporarily.

Some of the various sample test cases used in unit testing are: Test cases for path or branch covering (use structures basis testing method), Test cases for data flow coverage, Test

cases based on the experience (error guessing) such as: testing for boundary conditions- minimum, maximum and off by one boundary, testing with classes of bad data, too little data or no data, too much data, Wrong kind of data, Wrong size of data, Initialized data.

The unit testing for Neighbors Node Identification and Delay Calculation, Server Route Identification module has been carried out.

7.3 Integration Testing

Integration testing is a systematic technique for constructing the program structure, while at the same time conducting tests to uncover errors associated with interfacing. That is the program is constructed and tested in small segments, which makes it easier to isolate and correct.

The following common types of integration problems may be observed:

- Version mistakes.
- Data integrity violations.
- Overlapping functions.
- Resource problems especially in memory handling.
- Wrong type of parameters in function calls.

7.4 Validation Testing

Validation testing is the step where requirements established as a part of the software requirements analysis, are validated against the software already been developed. This test provides the final assurance that the software meets all the functionality, behavioral and performance requirements and rectifies the errors, which are uncovered during the coding.

Deviation or errors discovered at this step are corrected prior to the completion of the project with the help of user by negotiating to establish a method for resolving deficiencies. Thus, the proposed system under consideration has been tested by using validation testing and found to be working satisfactorily.

7.5 Output Testing

Asking the users about the format required by them tests the output generated or displayed by the system under considerations. Here the output format is considered in two ways. One is on the screen and another is on the printed format.

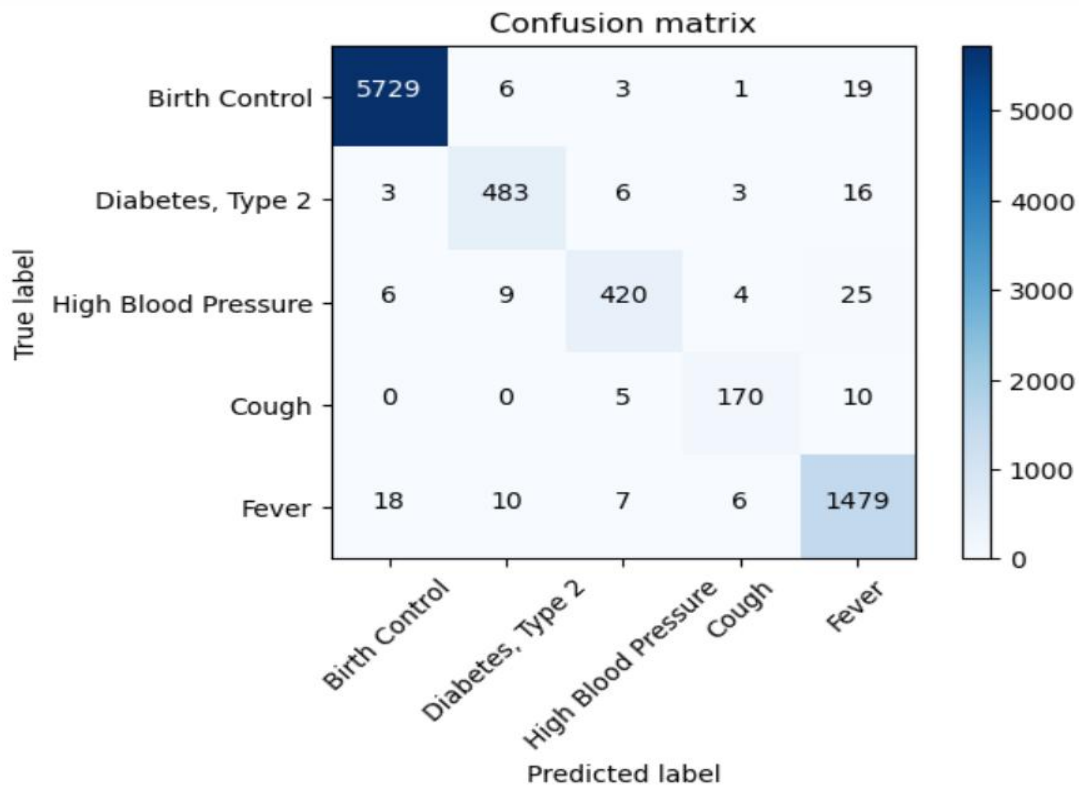


Fig 8.3 Machine Learning Model TFIDF

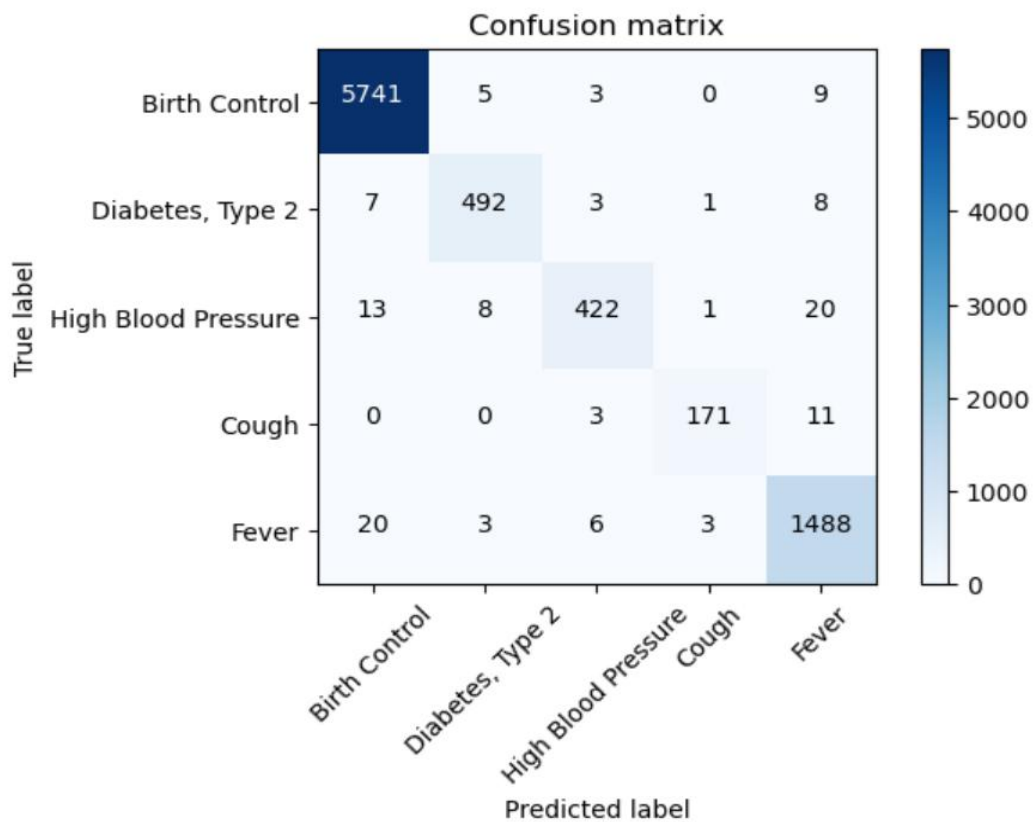


Fig 8.4 TFIDF: Bigrams

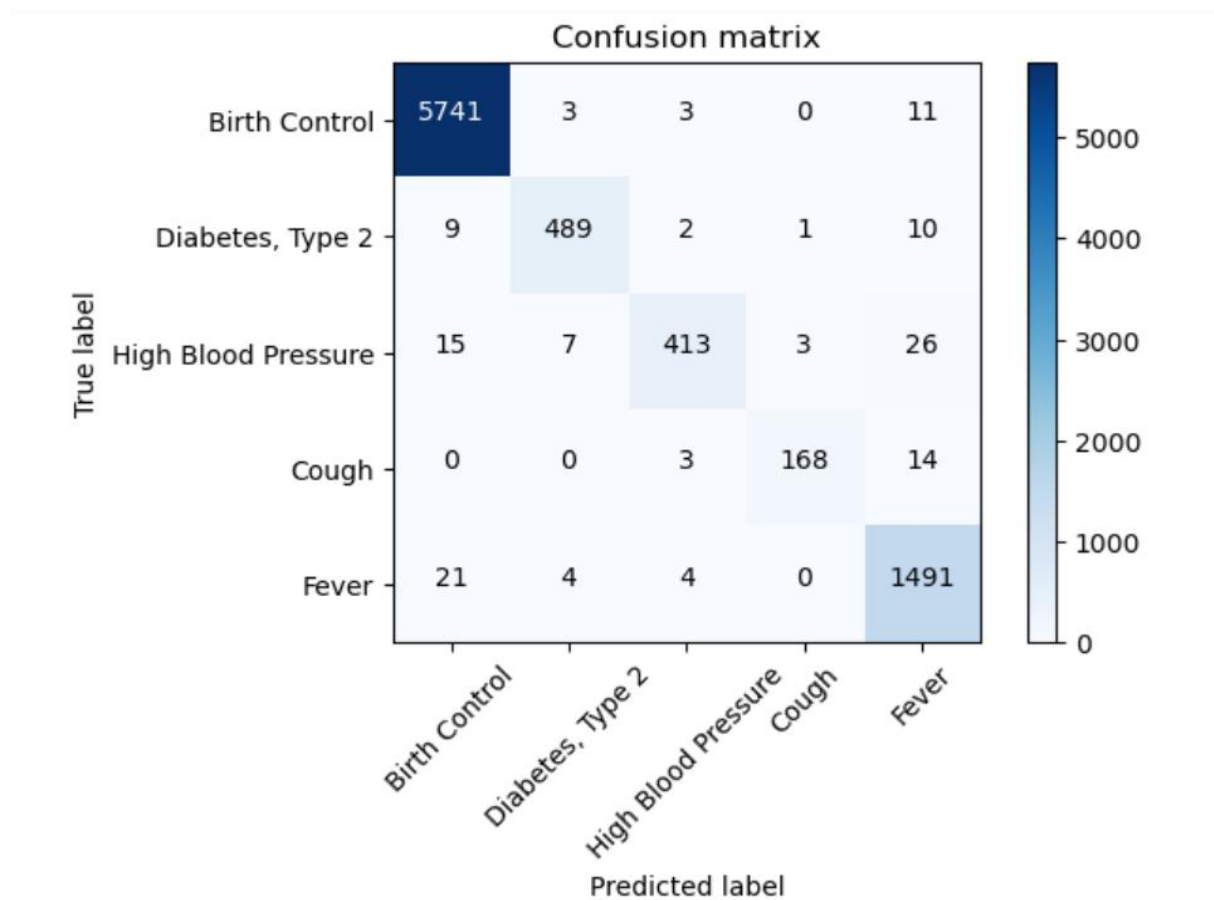


Fig 8.5 TFIDF: Trigrams

🔧 Patient Diagnosis and Drug Recommendation System 🔧

The system can detect the following diseases and recommend top drugs.

- [Acne](#)
- [ADHD](#)
- [Depression](#)
- [Diabetes_Type 2](#)
- [Migraine](#)
- [Pneumonia](#)

Enter Patient Condition:

Predict

User FeedBack:

feedback

Fig 8.6 User Interface

🔧 Patient Diagnosis and Drug Recommendation System 🔧

The system can detect the following diseases and recommend top drugs.

- [Acne](#)
- [ADHD](#)
- [Depression](#)
- [Diabetes_Type_2](#)
- [Migraine](#)
- [Pneumonia](#)

Enter Patient Condition:

Thirst and frequent urination are two classic diabetes signs caused by too much sugar in your blood. As your kidneys work harder to filter out the sugar, they also pull more fluids from your tissues, which is why you have to go to the bathroom more often than usual

Fig 8.7 Entering Patient Condition

Condition Predicted

Diabetes, Type 2

Top Recommended Drugs

Victoza

Glucophage XR

Invokana

Canagliflozin

Fig 8.8 Recommended Drugs

User FeedBack:

useful and i will recommend this

feedback

Feedback sent successfully!

Fig 8.9 User Feedback

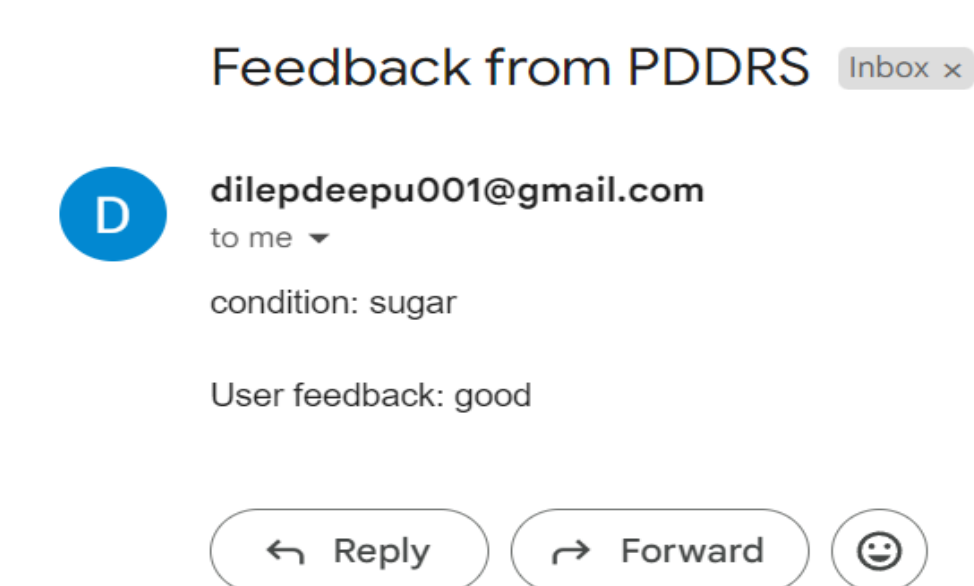


Fig 8.10 Feedback Sent to Mail

CONCLUSION

In this project we have successfully developed a drug recommender system that leverages patient reviews to provide medication suggestions. By utilizing advanced machine learning techniques and natural language processing (NLP), the system can predict the sentiment of patient reviews and recommend the most appropriate drugs for specific diseases. The use of various vectorization methods, such as Bag of Words (BoW), TF-IDF, Word2Vec, along with multiple classification algorithms, ensures analysis of patient feedback.

Key findings include TF-IDF vectorization achieved the highest performance, with an accuracy of 93%. The application of SMOTE effectively handled the issue of imbalanced data, improving the reliability of the machine learning models. The system's integration with a Streamlit app allows for user-friendly interaction, enabling healthcare professionals to input patient reviews and receive drug recommendations.

The email notification feature ensures that prediction results are promptly communicated to administrators, facilitating timely decision-making. The project demonstrates the potential of combining machine learning and NLP to enhance drug recommendation processes in healthcare, by reducing the workload of specialists and improving patient outcomes.

FUTURE ENHANCEMENT

While the current system shows promising results, there are several areas for further improvement and expansion:

Incorporating More Data Sources:

Expanding the dataset by including more diverse sources of patient reviews and clinical data can improve the system's accuracy and generalizability.

Advanced NLP Techniques:

Exploring more sophisticated NLP techniques, such as transformer-based models (e.g., BERT, GPT), could enhance the system's ability to understand and analyze patient reviews.

Real-time Data Processing:

Implementing real-time data processing capabilities would allow the system to provide up-to-date recommendations based on the latest patient feedback.

Integration with Electronic Health Records (EHR):

Integrating the recommender system with EHR systems can provide a more comprehensive view of patient health, leading to more personalized and accurate drug recommendations.

User Interface Enhancements:

Improving the user interface of the Streamlit app to include more features, such as detailed analytics and visualization of patient feedback trends, can enhance user experience and decision-making.

Clinical Trials and Validation:

Conducting clinical trials to validate the system's recommendations in real-world settings can help ensure its efficacy and safety.

Multilingual Support:

Adding support for multiple languages can make the system accessible to a broader audience and useful in different geographic regions.

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Certificate of Appreciation







